ASSESSING KNOWLEDGE

Assessing Access to and Application of Knowledge on Safer Building Practices in EQ Affected Districts

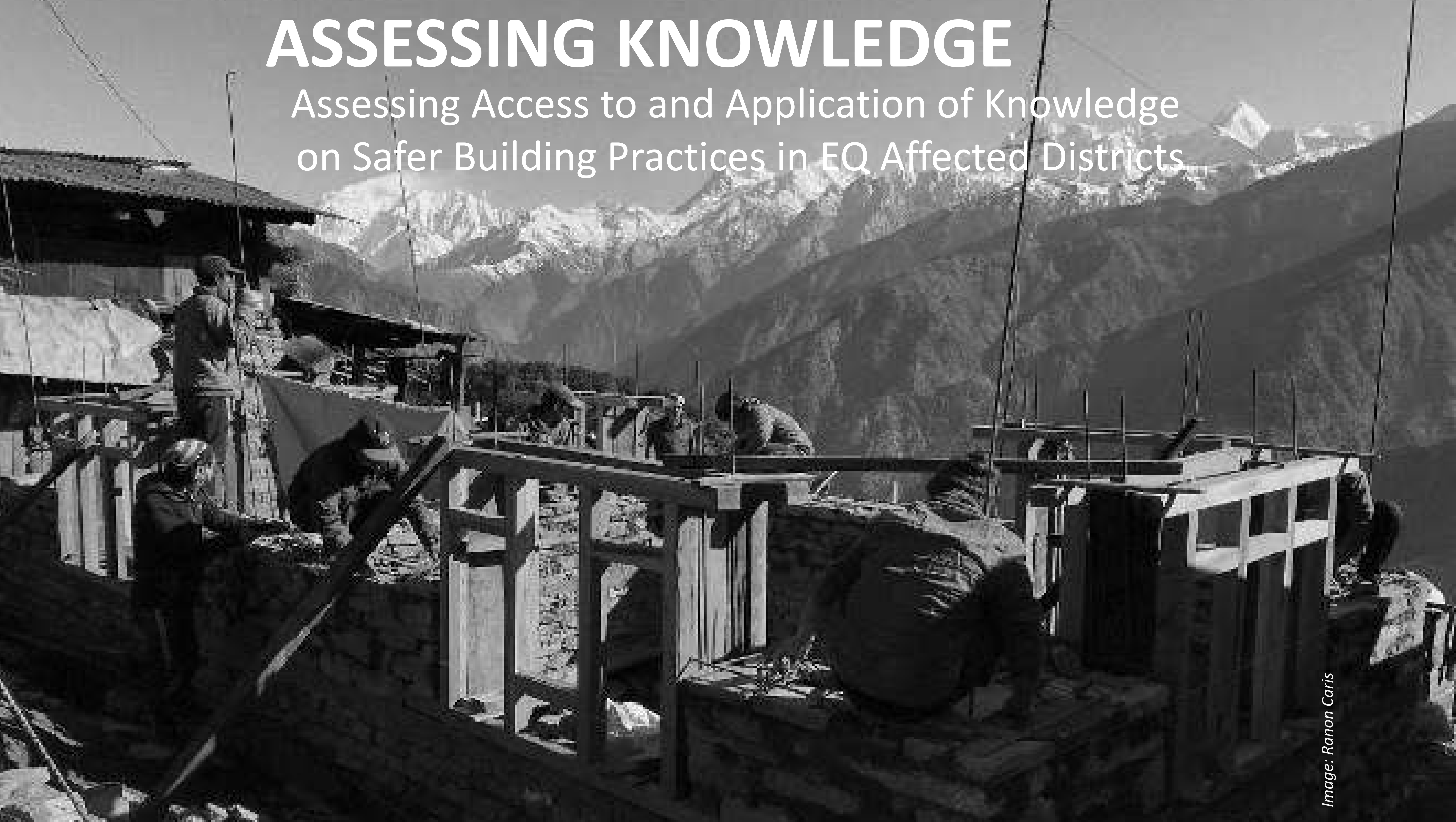


Image: Ranon Caris

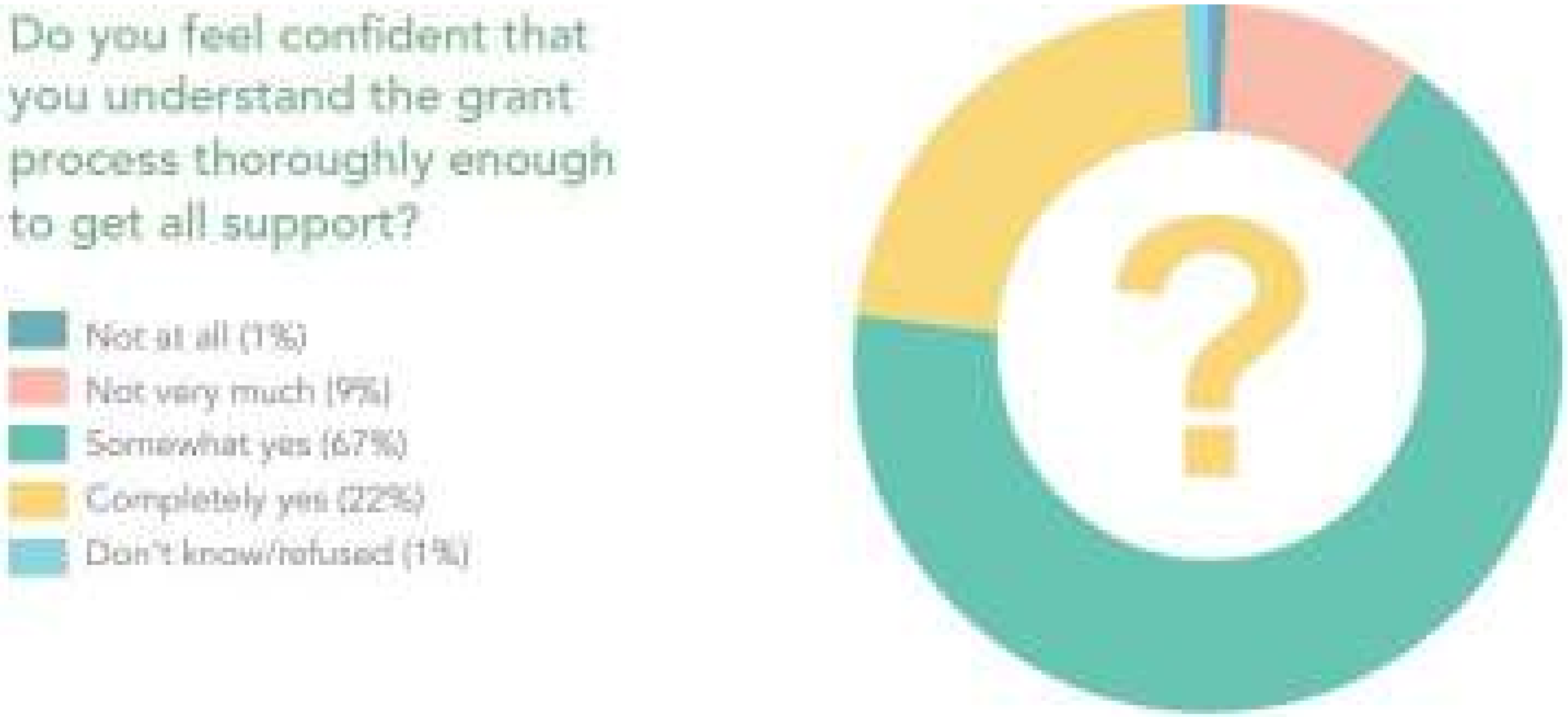
Community Perceptions – May 2018

ACCESS TO, AND APPLICATION OF, SAFER BUILDING PRACTICES

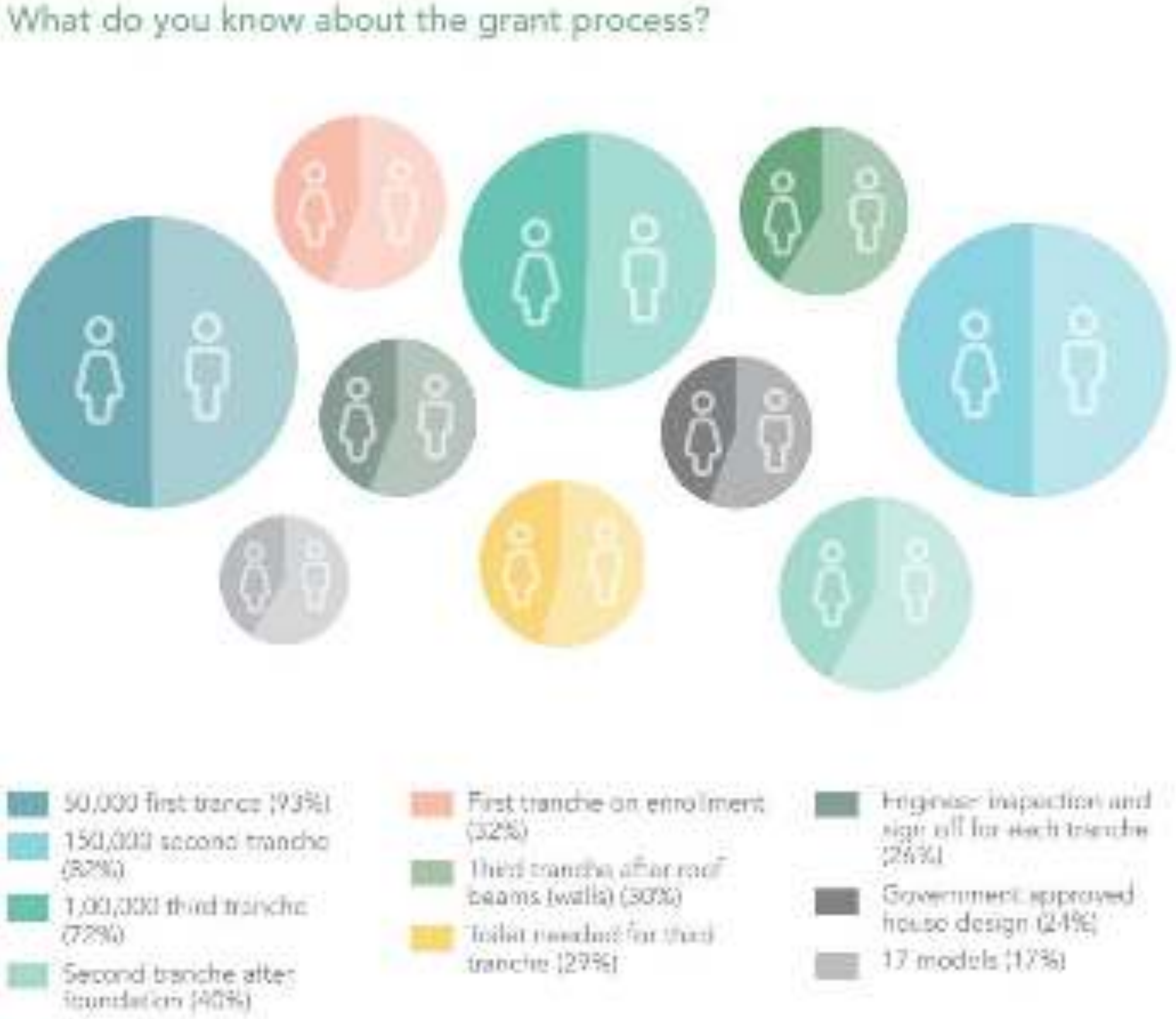
Community perception survey methodology

- All earthquake affected gaun/nagarpalikas above 50% damage
  - 118 G.P./N.P.s
  - Divided between three survey rounds, covering all 118 by end of year
  - Geographical spread in each round
- 31 G.P.s and 9 N.P.s every round
- 60 HH pre G.P., 80 HH per N.P.
- Total of 2580 respondents each survey
- 15 focus group discussions
- Feedback collected by 4 partner organizations
  - Care, DFID, BBC MA and HRRP

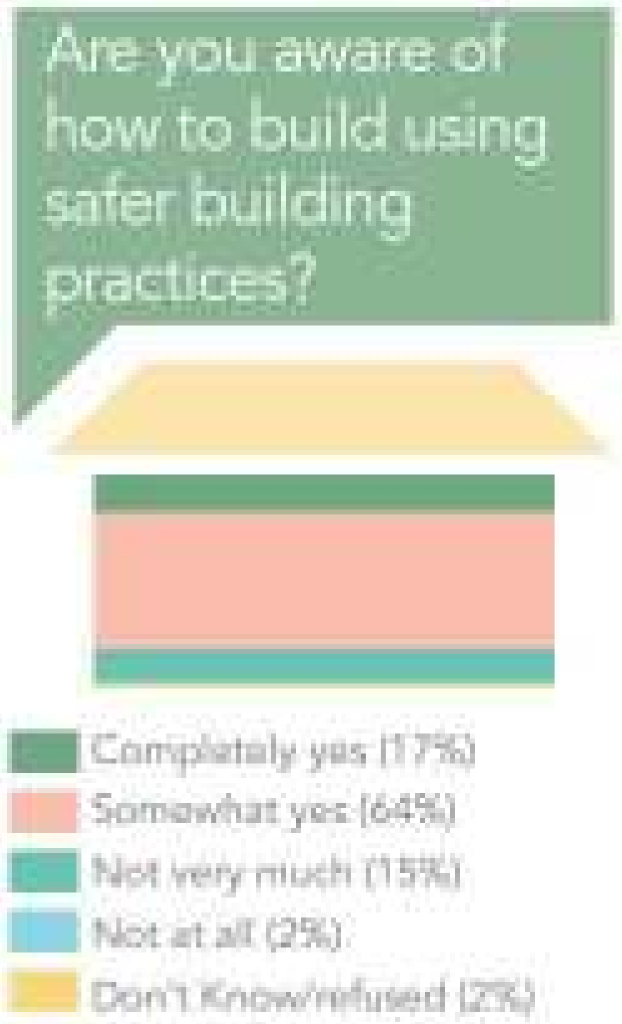
INFORMATION NEEDS



KNOWLEDGE OF GRANT PROCESS



KNOWLEDGE OF SAFER BUILDING PRACTICES



COMMUNICATION CHANNELS

SATISFACTION – HOUSE DESIGN

TWO ROOM HOUSES

DISTRICTS

Dolakha	Dhading
Kavre	Gorkha
Makwanpur	Sindupalchowk
Ramechhap	Solukhumbu
Rasuwa	Okhaldhung
Nuwakot	Sinduli

FEEDBACK SOURCE

Focus group discussion
Informal discussion

Many communities have flagged the concern that engineers will only provide them with the suggestion of building a two-room house, despite that size being insufficient for their lifestyle or family size. While not explicitly forced to select this model, communities report that engineers strong suggest that "it will be easy for them to approve" a two-room house, and they "will easily be able to move onto the next tranche" if they build a two-room house. This has left homeowners feeling like they are in the no-win situation of choosing between investing their limited resources in building a structure that will not meet their needs, or risk missing out on the government's grant.

"The two-room house which is widely suggested by engineers in our community is not appropriate for us. We don't know how to manage our physical assets, family members and guests in those two rooms. I think our community members are building that two-room house only for the sake of rebuilding. They continue to stay in their damaged house, and will stay further as well. In my opinion, to stay in such damaged house is a risk to their life."

– Dhading

SATISFACTION – TECHNICAL SUPPORT

Communities seek uniform and correct information from engineers so that they can utilize safer building practices and engineers can easily approve their construction within precise housing criteria. People want clear and authentic information related to reconstruction and government policies. However, many communities complain about varying information from one engineer to another. This creates confusion for homeowners. Even worse, in many instances homeowners who constructed based on the advice of one engineer are denied approval by another, leading to increase costs, wasted resources and significant delays in reconstruction.

INFORMATION GAPS

DISTRICTS

Dolakha Kavre
Makwanpur
Ramechhap
Rasuwa
Nuwakot
Dhading Gorkha
Sindupalchowk
Solukhumbu
Okhaldhung
Sinduli

FEEDBACK SOURCE

Focus group discussions
Information needs assessment

TECHNICAL GUIDANCE AND INSPECTIONS

DISTRICTS

Dolakha
Kavre
Makwanpur
Ramechhap
Rasuwa
Nuwakot
Dhading
Gorkha
Sindupalchowk
Solukhumbu
Okhaldhung
Sinduli

FEEDBACK SOURCE

Focus group discussions
Survey
Joint monitoring visits
Informal discussion
Information needs assessment

Irregular visits of government deployed engineers to assigned communities is resulting in people constructing their houses without consultation. This has created barriers to accessing the housing grant, as inconsistent technical guidance means houses often do not meet building requirements.

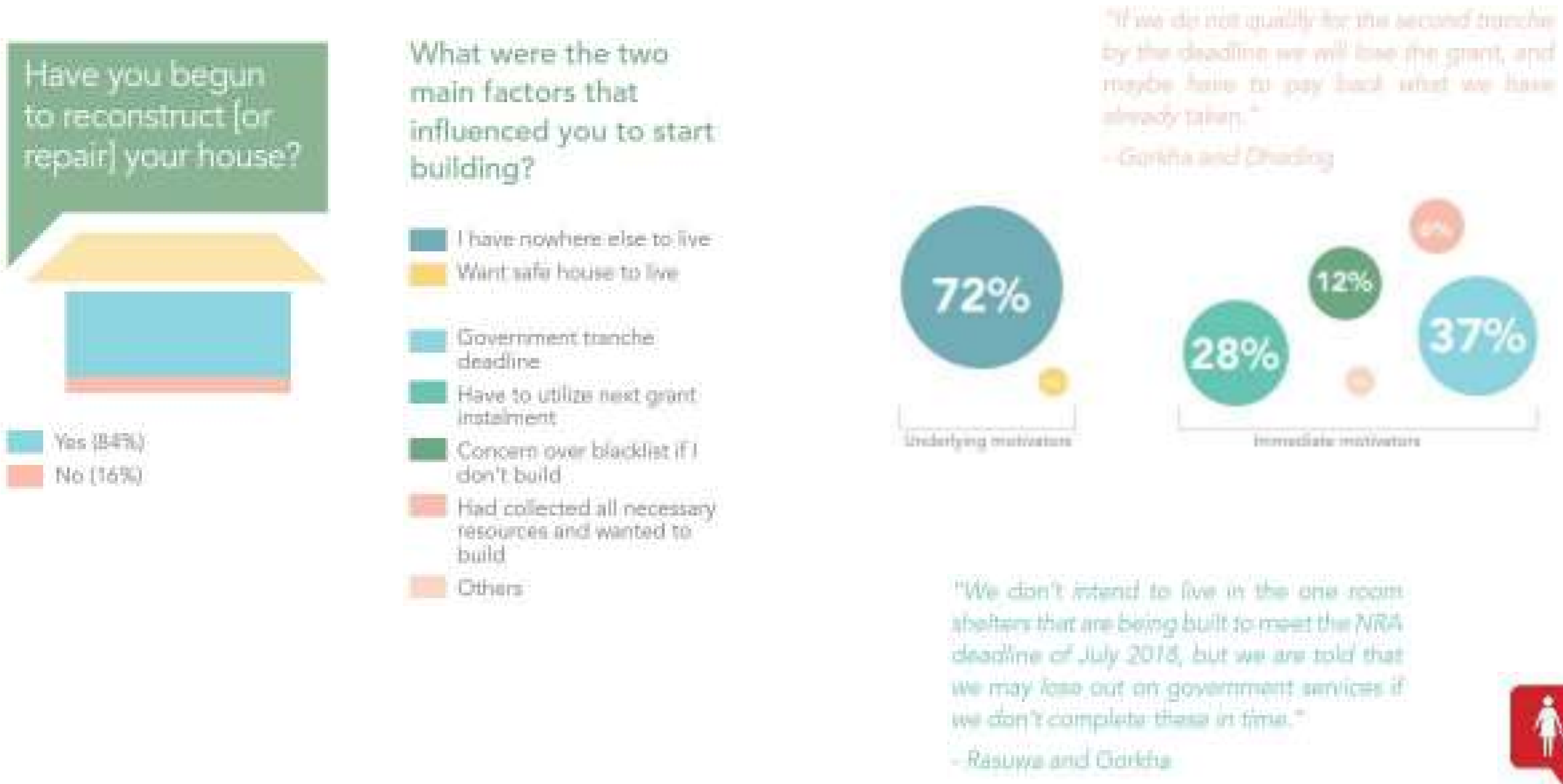
"The engineer suggested me to take a photo of every step of my home rebuilding process so that later he can them as evidence to approve the house. I did what he suggested. Now, he is refusing to approve, saying I did not build properly. It is such a stress for my family."

- Dhading

"Engineers do not come to inspect during ongoing construction. Even contacting them is difficult for us. Therefore, we have to collect information related to reconstruction through community mobilizers."

- Dhading

MOTIVATORS TO BUILD



Thank You

ASSESSING KNOWLEDGE
IN RECONSTRUCTION

Doctoral candidate MSc. Eefje Hendriks

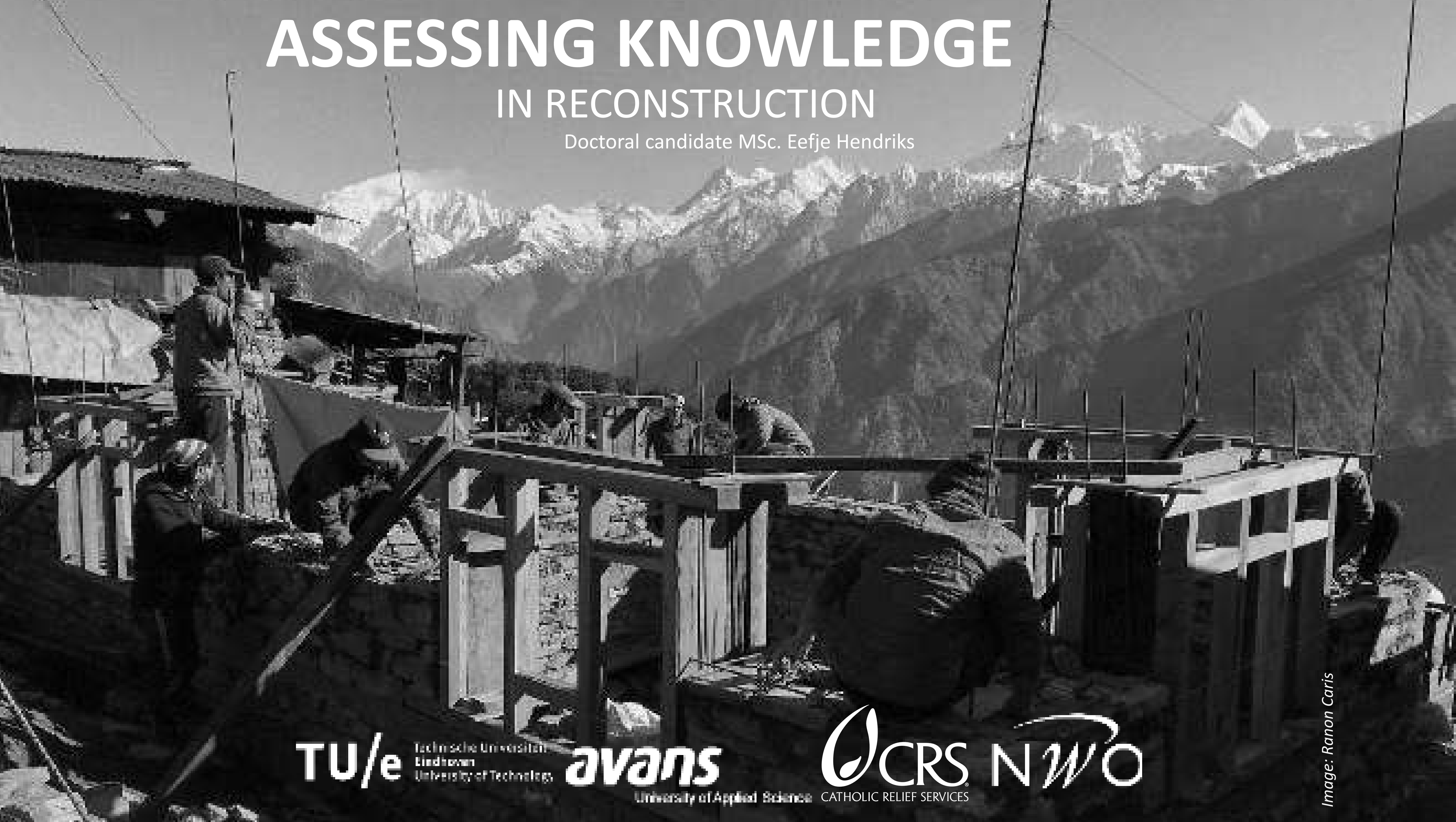


Image: Ranon Caris

The role of **knowledge** to enable **safer self-recovery**

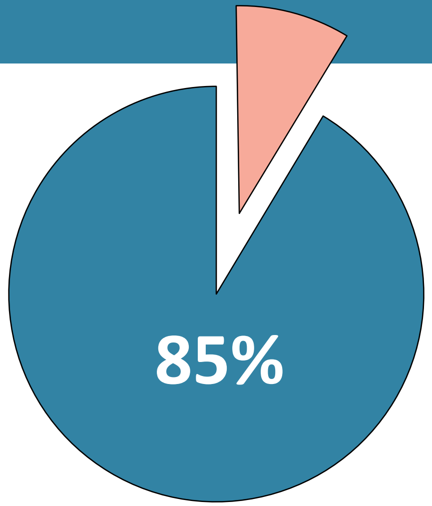


Image: Eefje Hendriks

Self-recovery ***+ knowledge*** *→ adoption → safer self-recovery*

Image: Tehelka

Understand **knowledge needs and application**

- Application of techniques to build back safer
- Understanding and awareness of earthquake resistant construction techniques
- Barriers and drivers to build back safer
- Needed and provided knowledge
- Preferred and trusted knowledge sources

Research tools

Household questionnaire

Focus groups

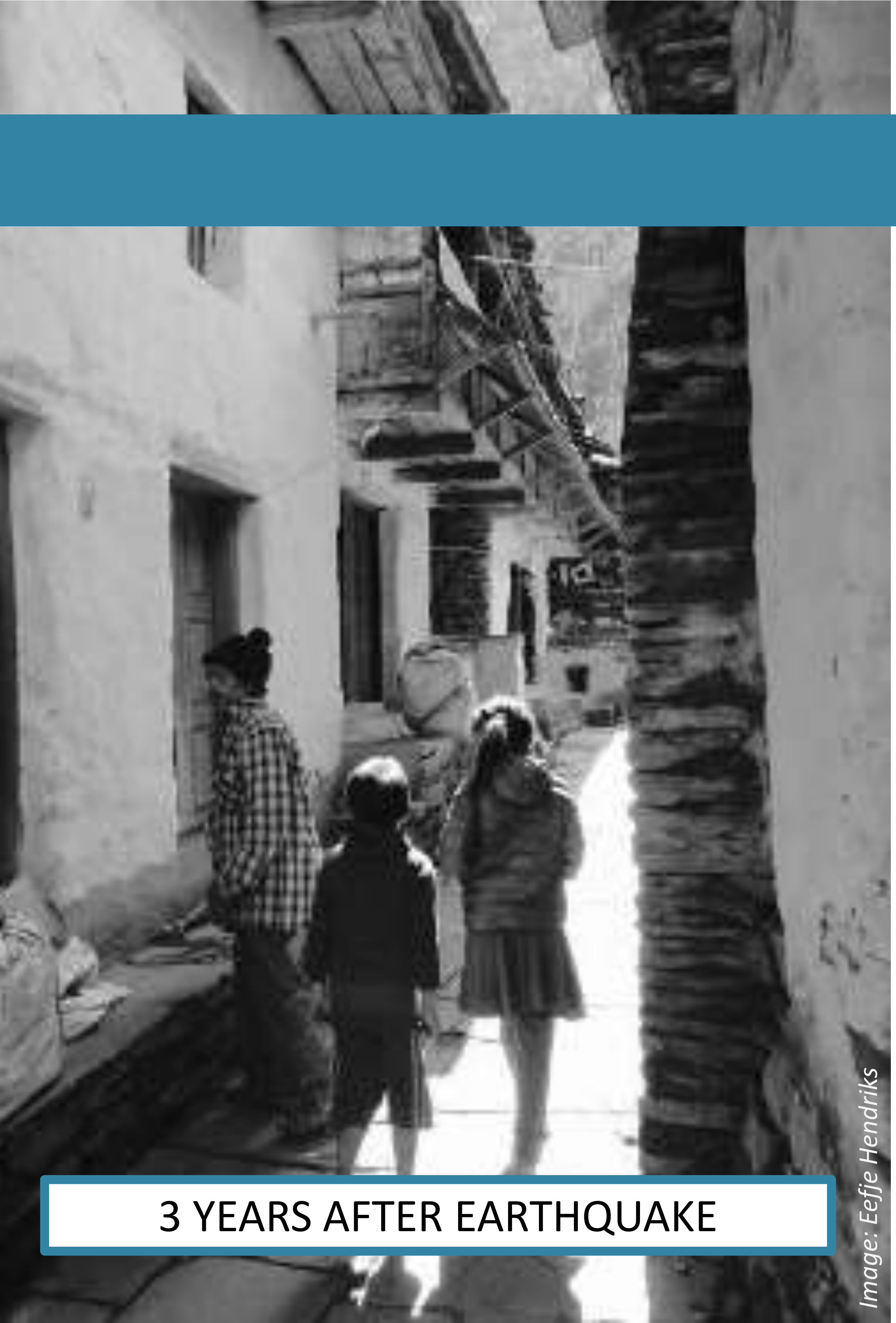
Structural assessments

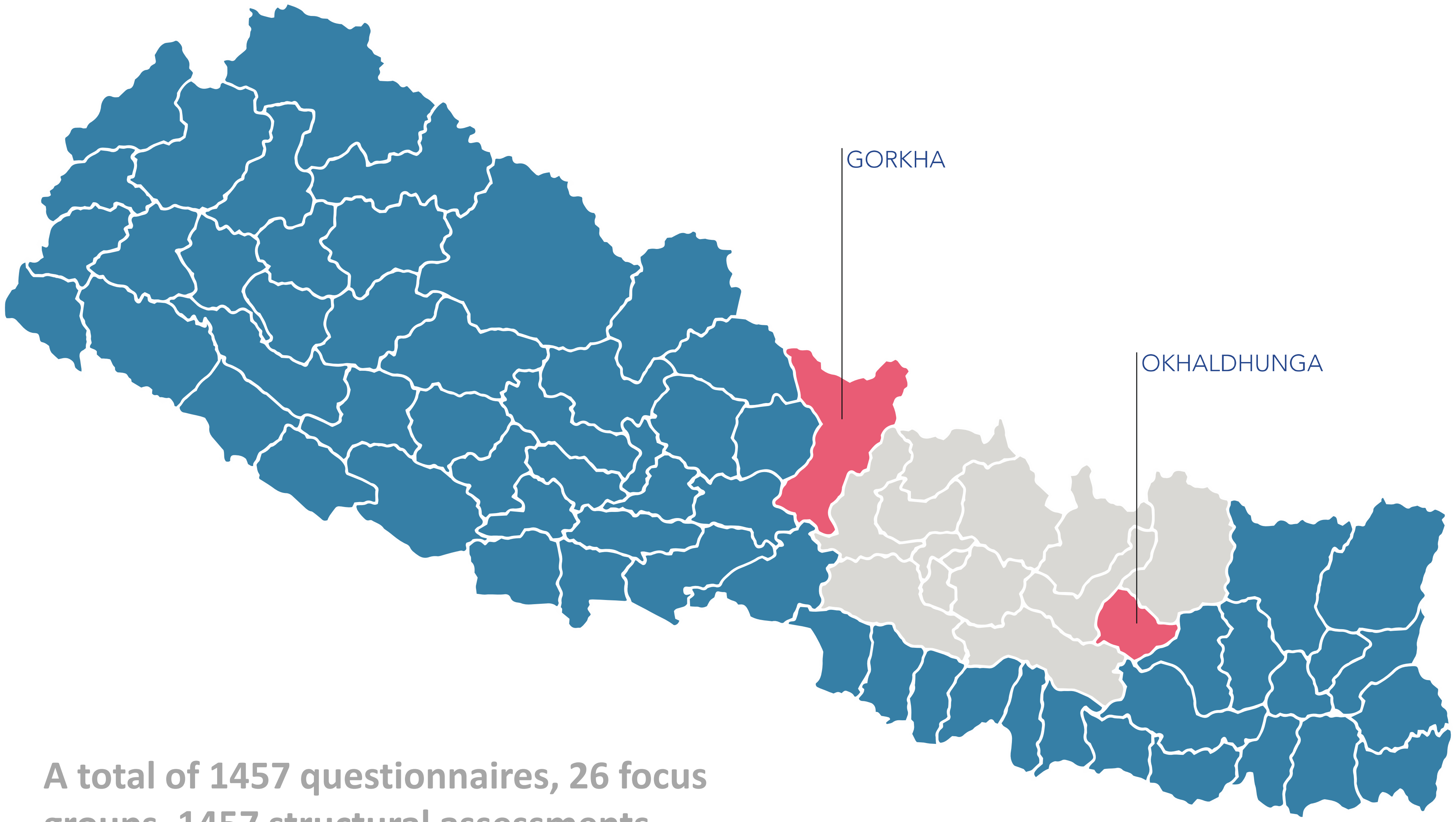
Key-stakeholder interviews

Ward data

Participatory construction

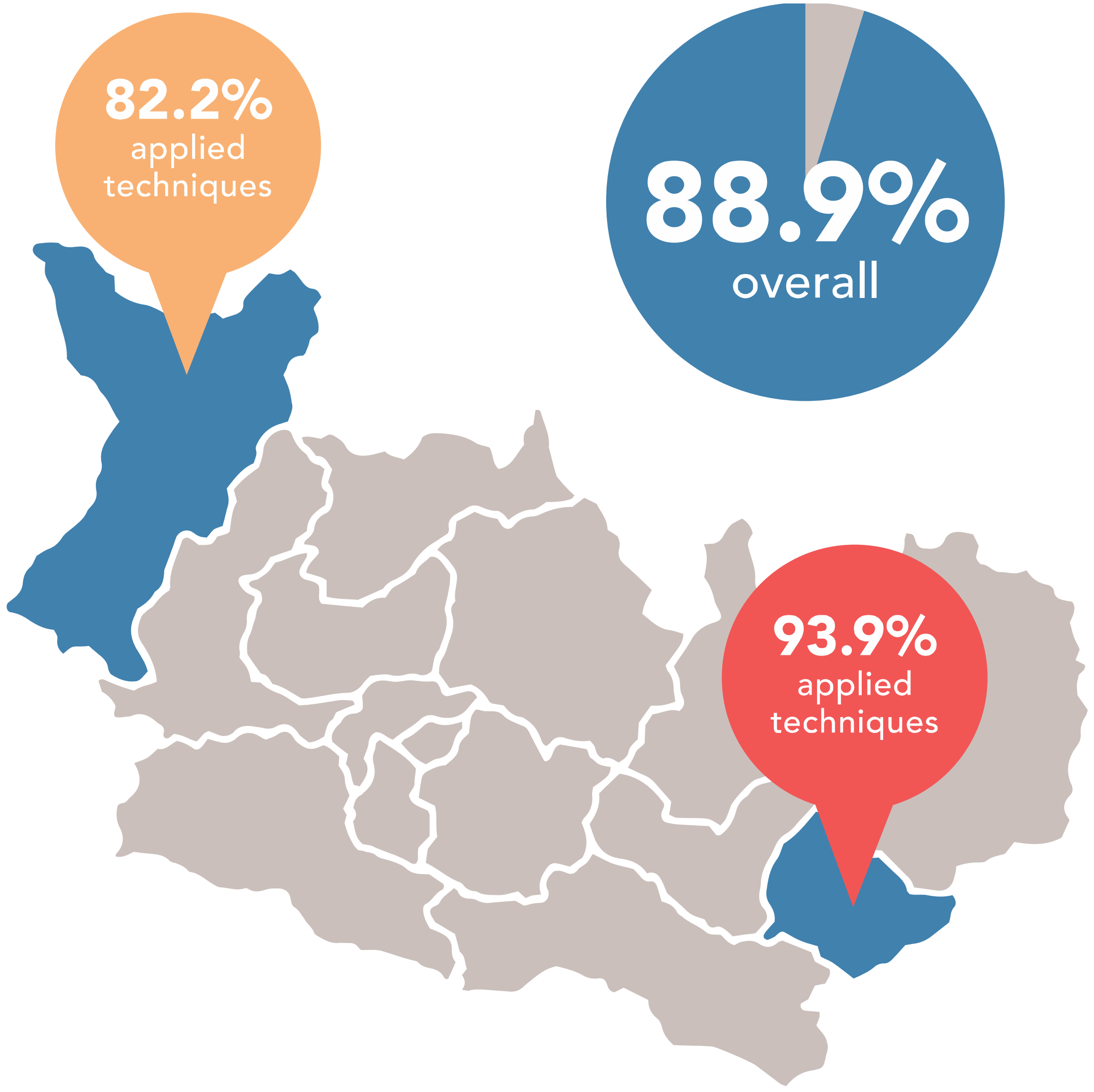
Research sample





A total of 1457 questionnaires, 26 focus groups, 1457 structural assessments and 70 key-stakeholder interviews were conducted in 26 wards in 8 VDC's in Gorkha and 14 VDC's in Okhaldhunga.

Application of techniques to build back safer



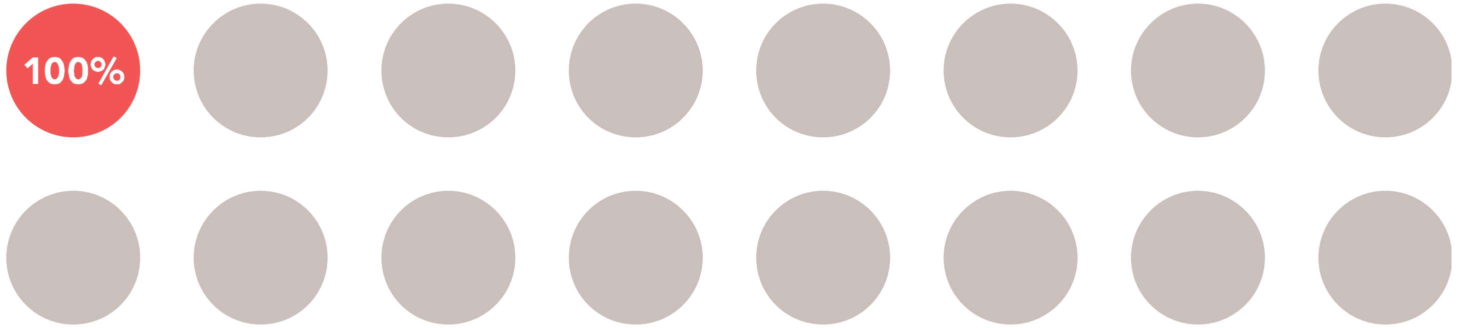
GORKHA OKHALDHUNGA

GORKHA

4 / 8

wards applied the techniques 100%

OKHALDHUNGA



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wards applied the techniques 100%

Did you apply earthquake resistant techniques?

Application of techniques to build back safer

	GORKHA	OKHALDHUNGA
Bands	1 95.7%	1 80.9%
Reinforced concrete	37.9%	24.5%
Proper distance between openings and corners	3 53.5%	9.6%
Floor tied to the walls	2 65%	11.9%
Roof fixtures, tied down	43.5%	10.9%
Cross-bracings used	28.4%	10.6%
DPC used	0.8%	3 65.4%
Bigger foundation	0%	54.7%
Deeper foundation	1.3%	2 72.5%

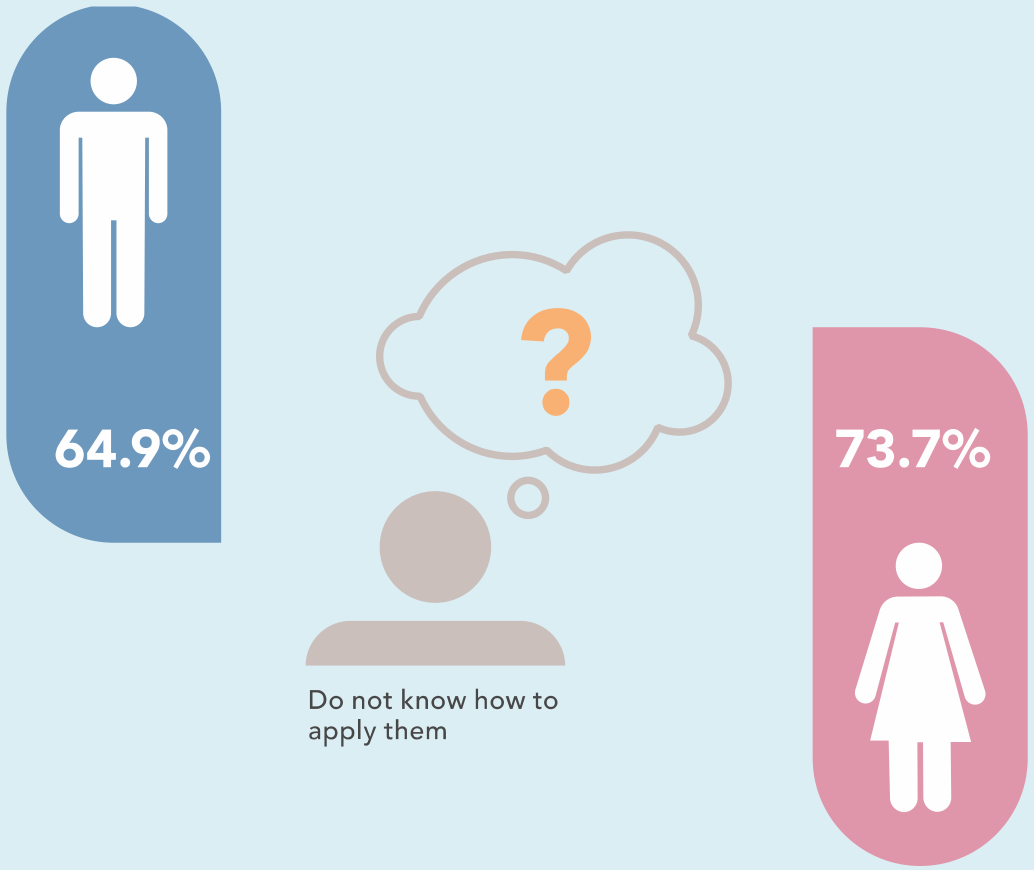
What kind of earthquake resistant techniques did you apply?

GORKHA   OKHALDHUNGA

Application of techniques to build back safer

93.3% G 23.1% O
Did not know how to apply them

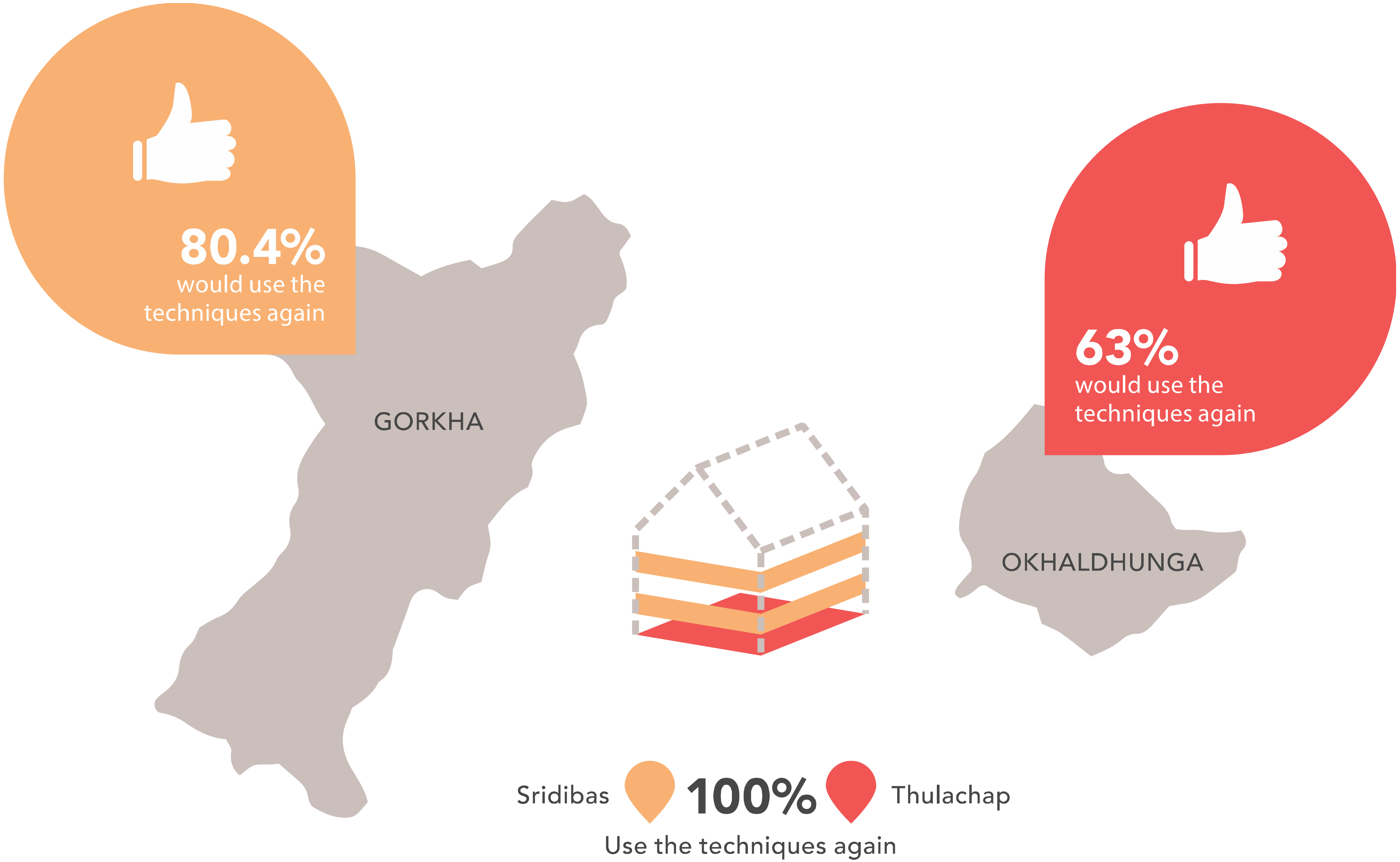
4.0% G 17.9% O
There is no need



Why did you not apply the techniques?

G GORKHA O OKHALDHUNGA

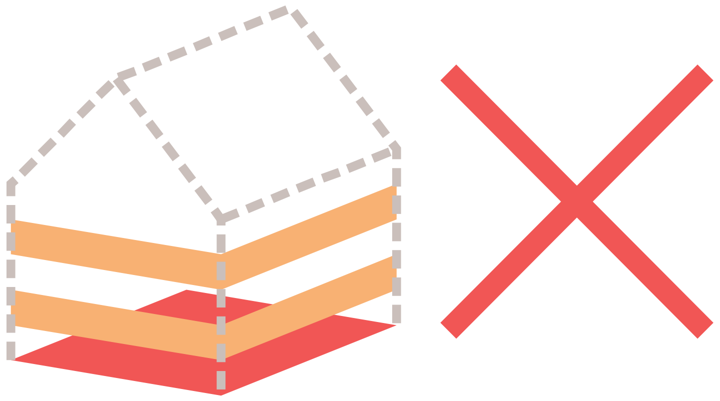
Application of techniques to build back safer



Would you apply earthquake resistant techniques in a future construction project?

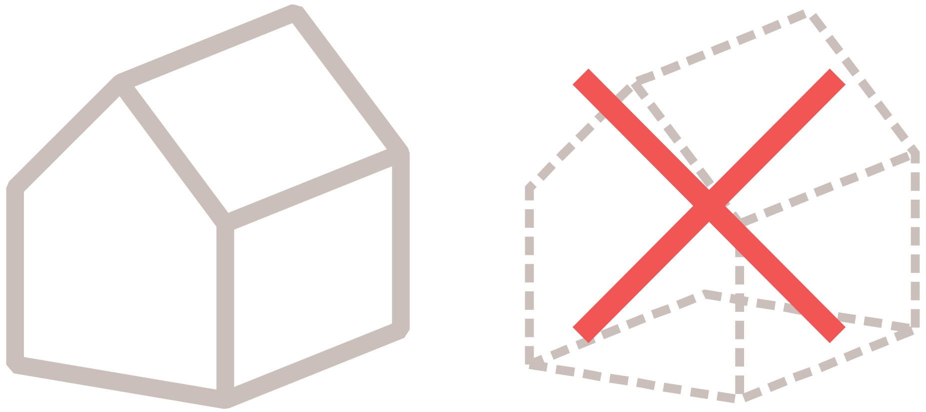
Application of techniques to build back safer

Would you NOT apply earthquake resistant techniques in a future construction project?



83.3% G **57.9%** O

think there is no need for earthquake resistant techniques

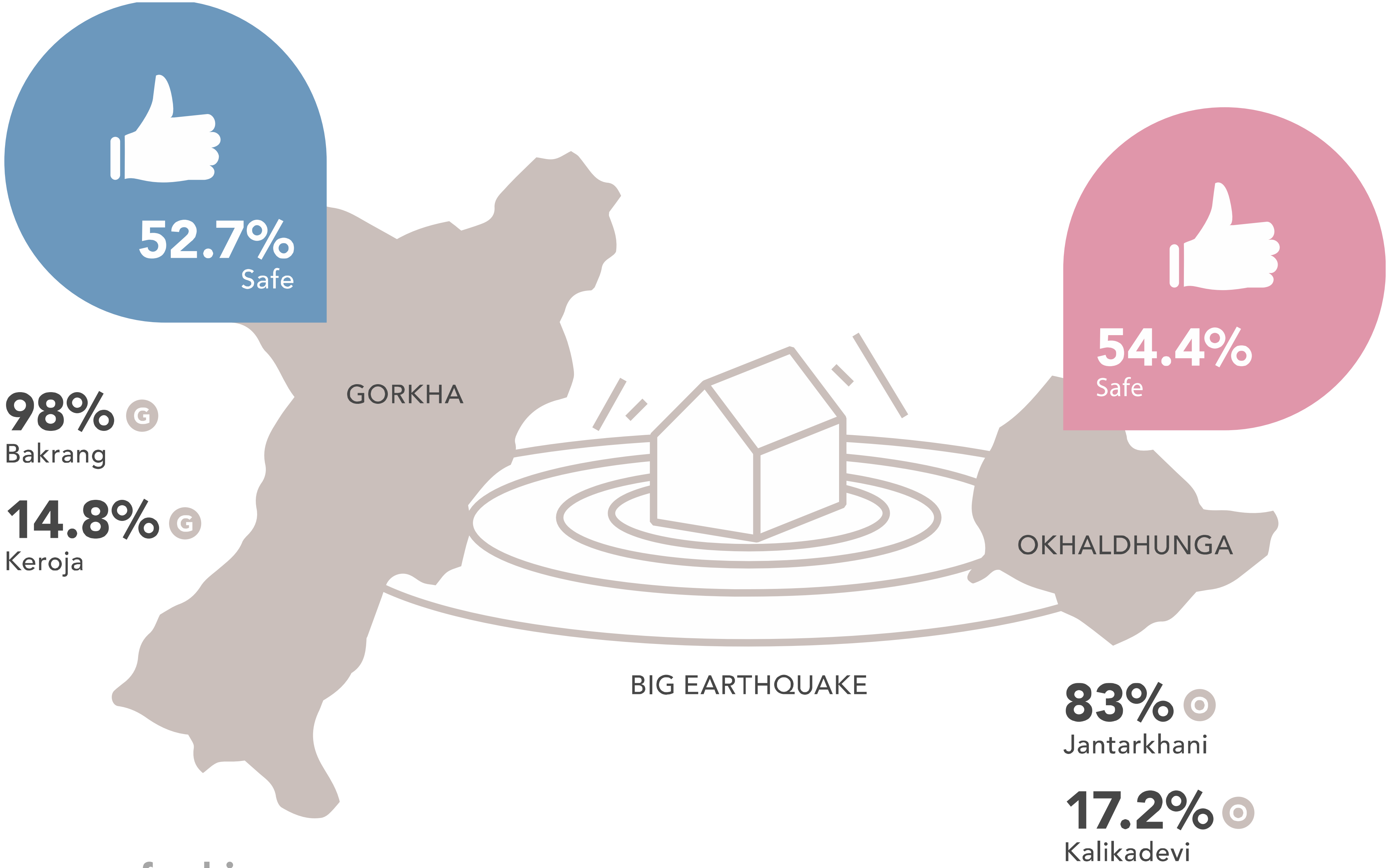


29.6% O

are not going to build a house again

100% Bakrang Gyalchowk

Awareness of constructive risks



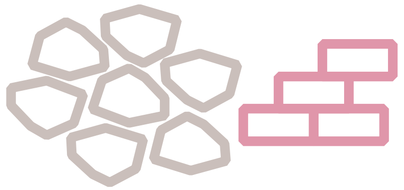
Do you feel safe in case of a big earthquake?

G GORKHA O OKHALDHUNGA

Awareness of constructive risks

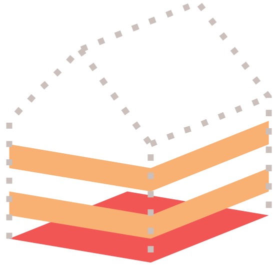
What makes your house safe in case of an earthquake?

39.2%
applied principles
to construct
earthquake resistant

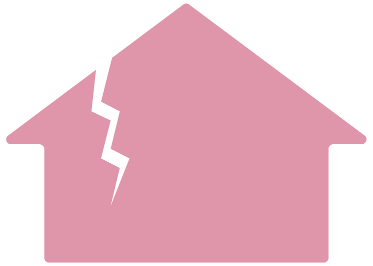


24.6% G
19.5% O
Strong materials

21.5% G
27.9% O
Engineer's advice



61.5% G
24.3% O
Earthquake resistant
constructin principles

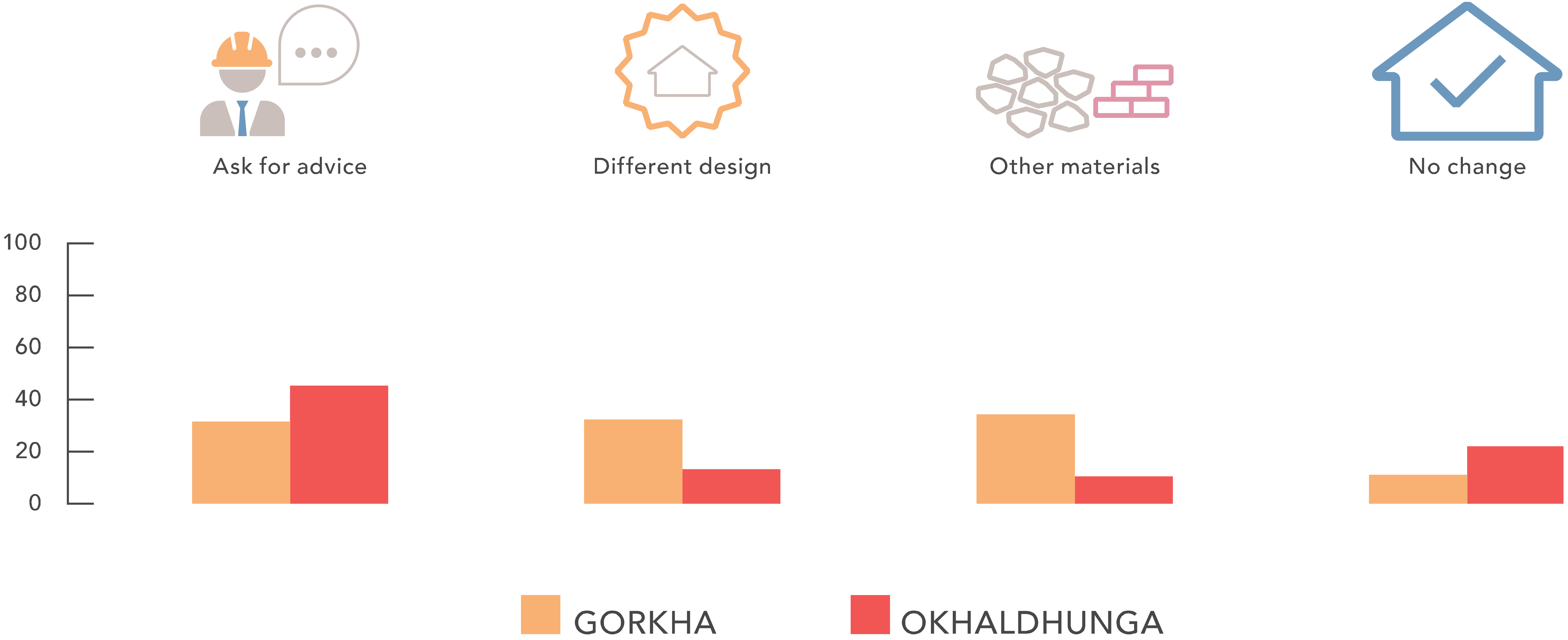


8.6% G
17.5% O
House is not strong

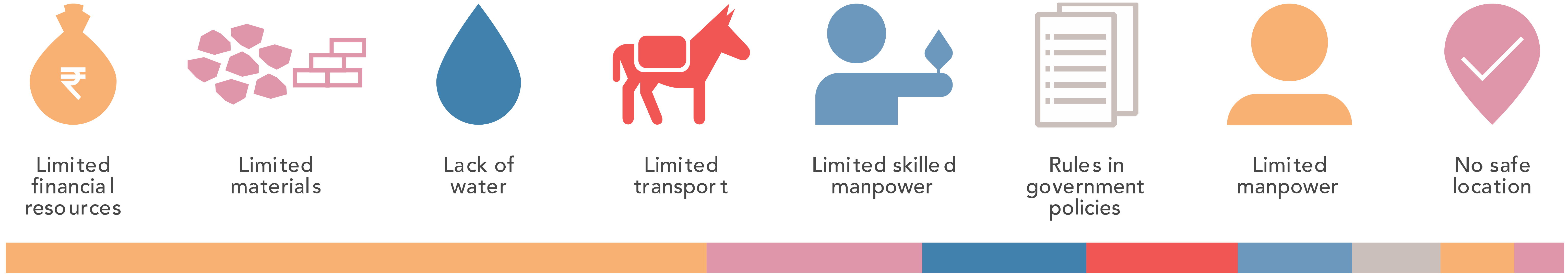
G GORKHA O OKHALDHUNGA

Understanding of constructive risks

If you would have to build a house again what would you do different?



Barriers for good construction practice



What limited you to build back an earthquake resistant house?

Drivers for good construction practice

27%
Assistance by Government of Nepal

15%
General need for safety

12.3%
The earthquake

6.9%
Assistance from construction professionals

6.7%
Availability of knowledge to build back safer

5.8%
Assistance for shelter

5.4%
General need for shelter

Drivers for good construction practice

Drivers being mentioned

The earthquake
Gorkha **50%**
Okhaldhunga **96%**

Availability of knowledge to build back safer
Gorkha **50%**
Okhaldhunga **75%**

Organisations
Gorkha **50%**
Okhaldhunga **31%**

Fear of the blacklist
Gorkha **20%**
Okhaldhunga **6%**

Needed knowledge

What question did you have related to the reconstruction of your house?

33.2% G
Where to find good mason

24.4% O
Advice wanted from engineer

62.8% G **28.6%** O
Materials

24.8% G
How to construct in general

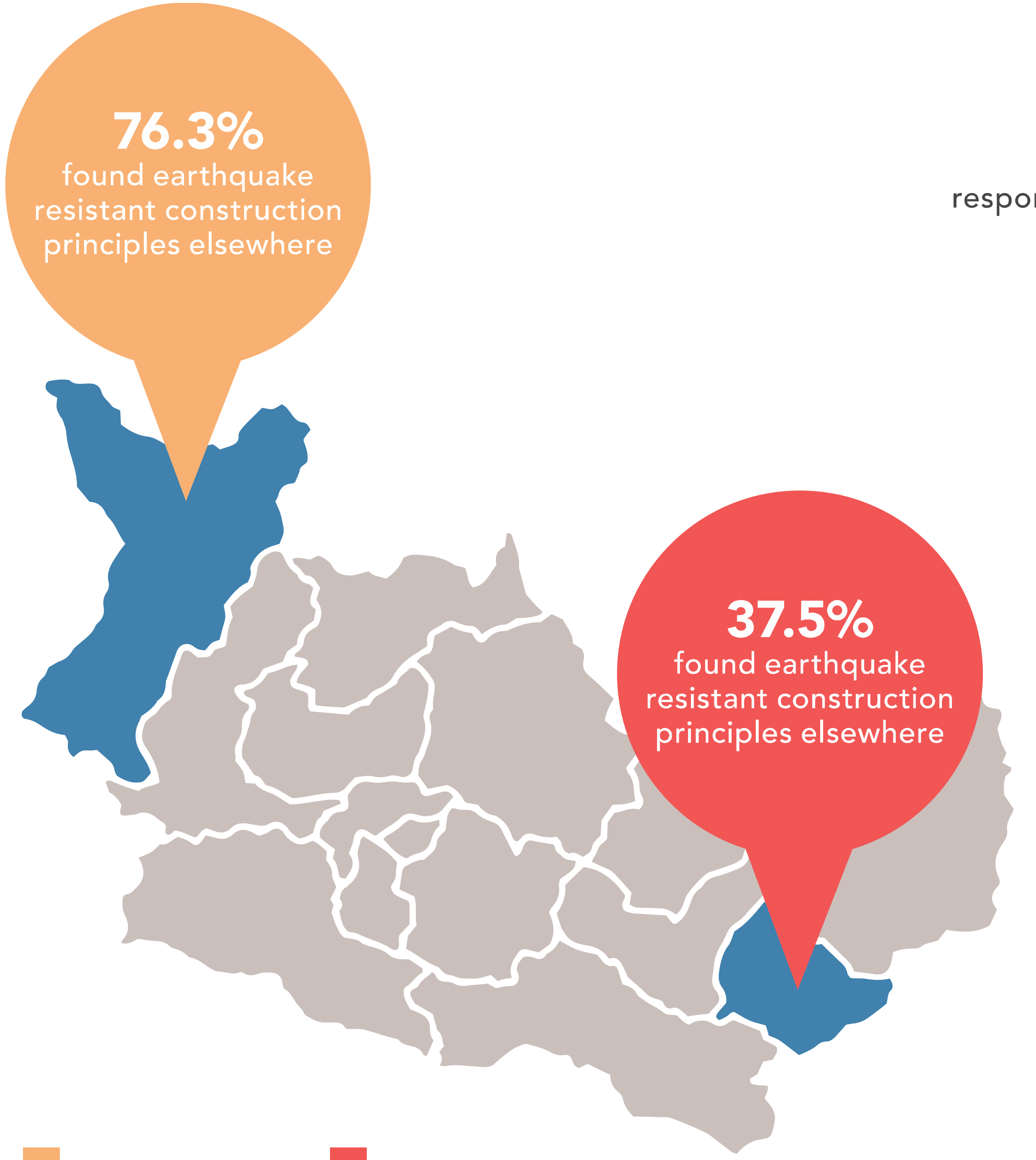
59.7% G **26.2%** O
Safe materials

36.7% G
How to construct earthquake resistant

46% G
What foundation to build

G GORKHA O OKHALDHUNGA

Needed knowledge



GORKHA OKHALDHUNGA

more than 50%

respondents have not found earthquake resistant construction principles elsewhere in

Harkapur Jantarkhani Kalikadevi

Kaptigaun Khiljifalate Raniban

If you did not receive training, did you find such information somewhere else?

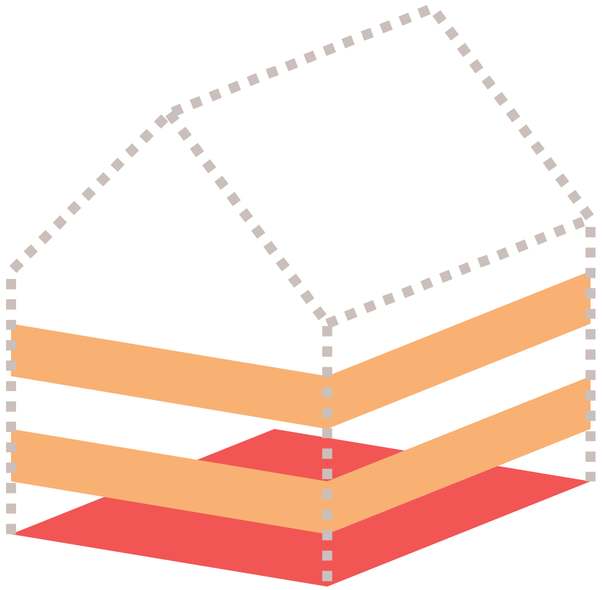
Needed knowledge

29.5% O
In the neighbourhood

41.1% O
Local contractor

17.9% O
Copy from other houses

20.2% G
VDC training



24.2% O
VDC engineer

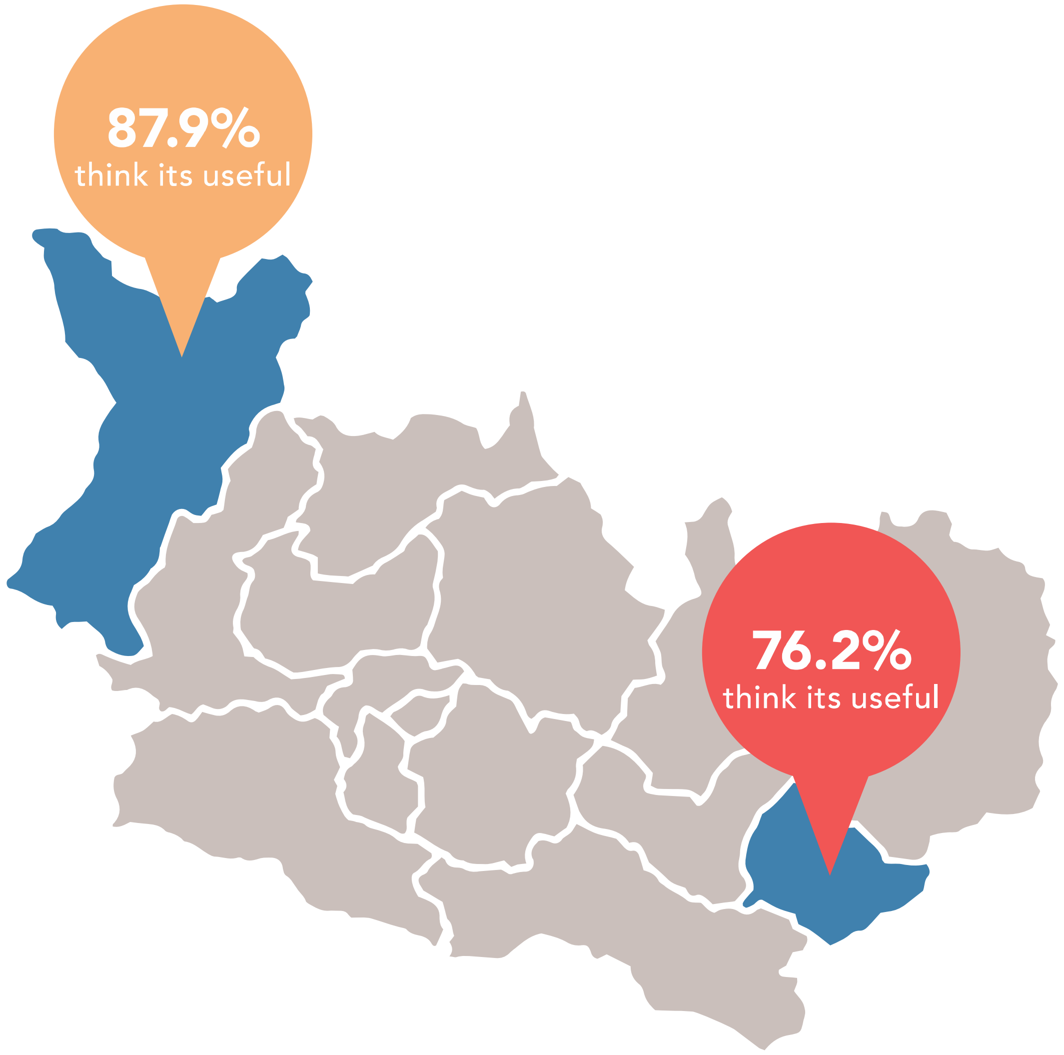
13.1% O
Training

14.1% O
Trained person/
experienced worker

Where did you find these techniques?

G GORKHA
O OKHALDHUNGA

Provided knowledge



GORKHA OKHALDHUNGA

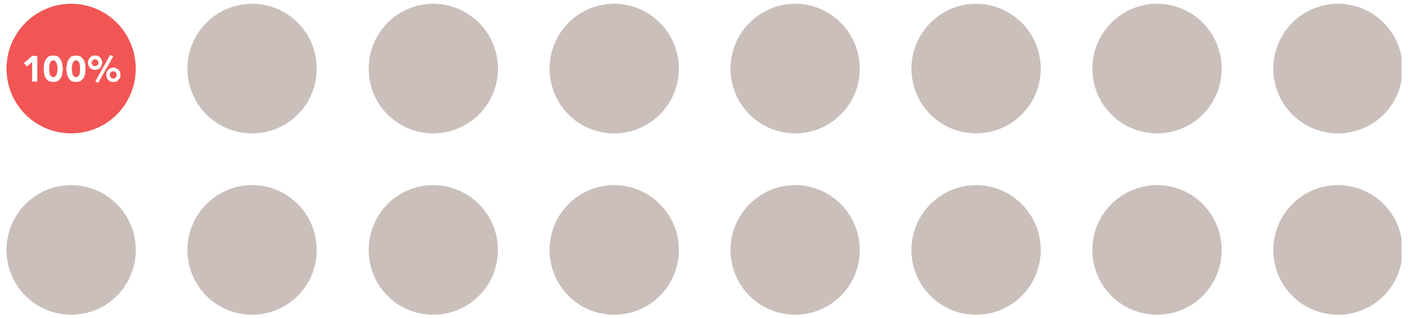
Did you think the training was useful?

GORKHA

4 / 8

wards think training was 100% useful

OKHALDHUNGA



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wards think training was 100% useful

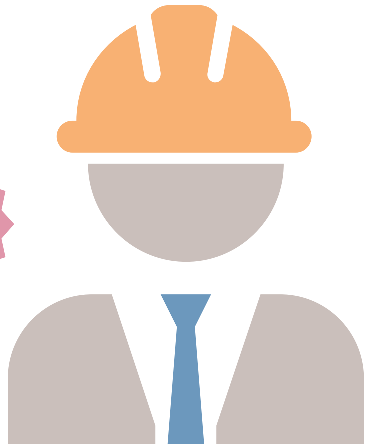
Preferred knowledge sources

1

28.5%
Construction professionals

18.5%
Engineers

2



Actors supporting to build back safer

Knowledge sources to build back safer

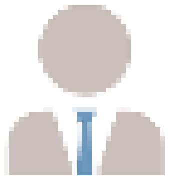
	GORKHA	OKHALDHUNGA
Community meeting	2	1
Radio		2
Demo house	1	
Door to door	3	
Training		
TV		3
Engineer advice		3

GORKHA
OKHALDHUNGA

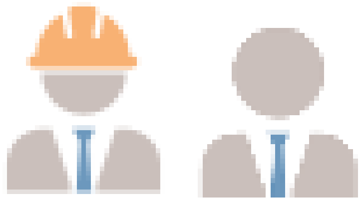
Key-stakeholders improvement of their role

Findings:

- Most engineers and contractors want to improve their role as knowledge source by ***receiving more training***



- The ward leaders want to improve their role as knowledge source by both consulting the ***local engineers*** and ***higher government***.



- Both engineers and ward leaders consider ***the District Office*** as a source of information for improvement of their role.

- The contractors and ward leaders consider the ***local engineers*** as an important source of information for improvement of their role.

Potential for adoption during post-disaster recovery

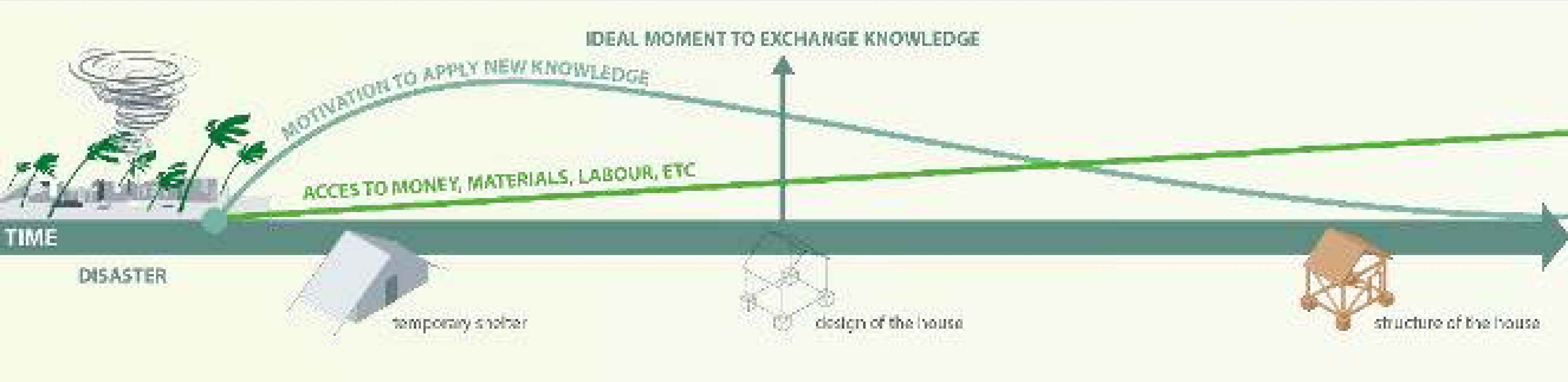
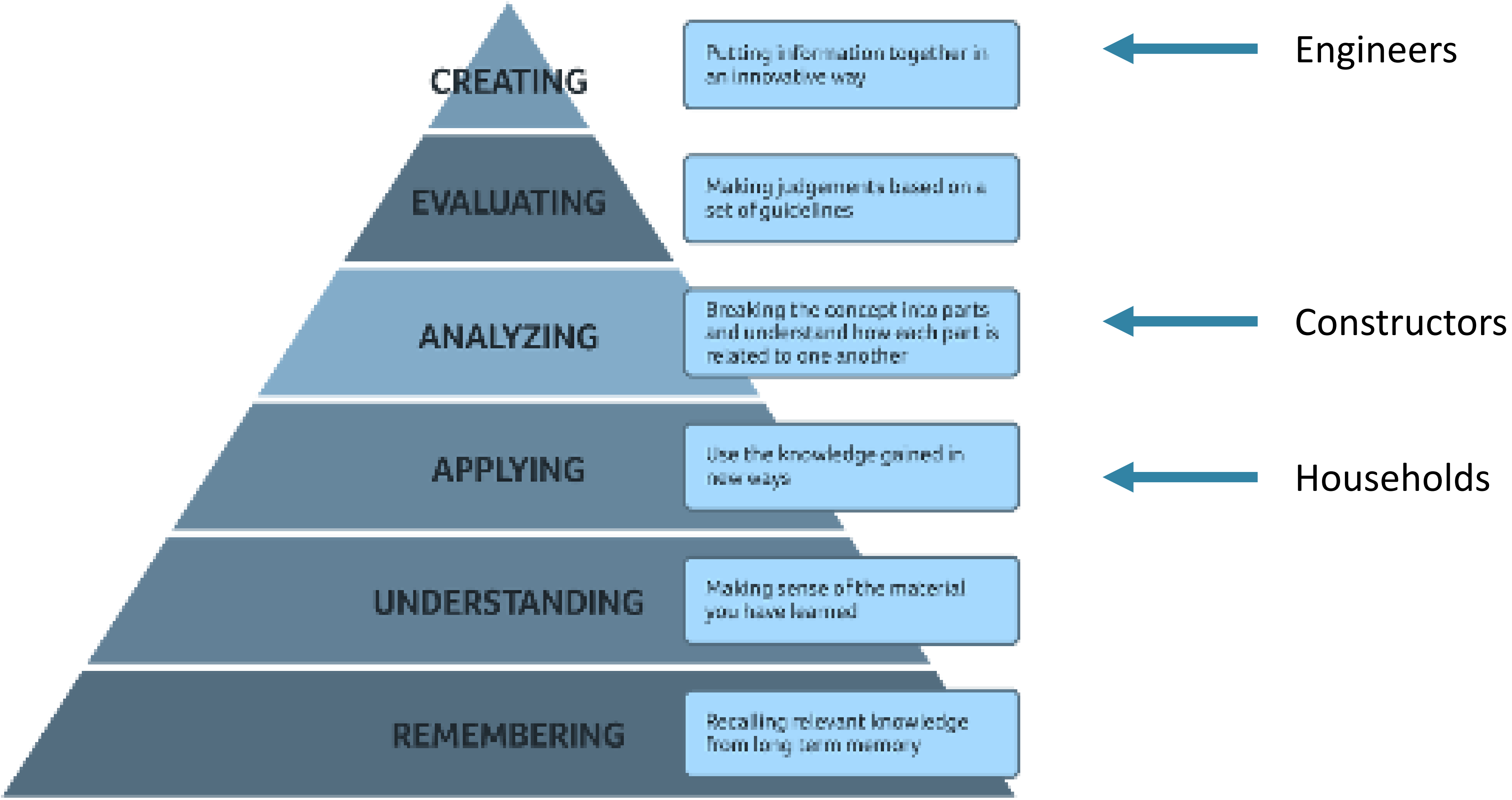


Image: Eefje Hendriks

3 YEARS AFTER EARTHQUAKE

Key-stakeholders improvement of their role



Debate

Statements:

- People have **the right to correct information** being provided to them about deadlines and blacklists. (Siobhan)
- Engineers in the field do not have the expertise to **inspect alternative designs** that are earthquake resistant and fit the homeowners needs. (Smriti / Kshitij)
- Okhaldhunga is an example of **best practice** for successful self-recovery and should be replicated in the rest of Nepal. (Eefje)

Solutions

Solutions:

- How can the **motivation** to build earthquake resistant be generated and maintained in the future inside and outside the earthquake affected districts? (Siobhan)
- What methods for **knowledge exchange** can be used to increase satisfaction and safety? (Smriti / Kshitij)
- What is needed to **improve comprehension and then the communication** by socio technical assistance field staff? (Eefje)